

Reproducing “Bilinear MLPs Enable Weight-Based Mechanistic Interpretability”

FACT-AI Course Project

January 28, 2026

Bilinear MLPs: Weight-Based Interpretability

Purpose:

- ▶ Direct interpretability from weights
- ▶ No post-hoc analysis needed
- ▶ No SAE training required

Structure:

$$\mathbf{y} = (\mathbf{W}_l \mathbf{x}) \odot (\mathbf{W}_r \mathbf{x})$$

Benefits:

- ▶ Element-wise multiplication creates quadratic interactions
- ▶ Interaction tensor can be eigendecomposed
- ▶ Interpretable features directly from weights

Eigendecomposition for Interpretability

Method:

1. Extract interaction tensor for each class c :

$$B_c = \mathbf{W}_l \otimes \mathbf{W}_r$$

2. Symmetrize the tensor:

$$B_c^{\text{sym}} = \frac{1}{2}(B_c + B_c^\top)$$

3. Eigendecomposition:

$$B_c^{\text{sym}} = \sum_{r=1}^d \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^\top$$

Effective Rank:

$$\text{EffRank}(\lambda) = \left(\frac{\|\lambda\|_1}{\|\lambda\|_2} \right)^2$$

What Went Well: Vision Results

MNIST (Full Regularization: $\sigma=0.5$, $\lambda=1.0$): Top Eigenvectors

Key Result:

- ▶ Effective rank: 15.5 (WD) vs 38.5 (none)
- ▶ Ratio: $0.40 < 0.5$ ✓
- ▶ Accuracy: 97–98%

Decomposition Formula:

$$B_c^{\text{sym}} = \sum_{r=1}^d \lambda_r^{(c)} \mathbf{v}_r^{(c)} (\mathbf{v}_r^{(c)})^T$$



Figure: Regularized eigenvectors

Vision Results: Key Figures

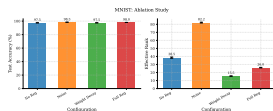


Figure: Accuracy vs Effective Rank

$$B_c^{\text{sym}} = \sum \lambda_r \mathbf{v}_r \mathbf{v}_r^T$$

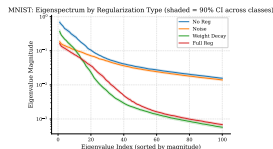


Figure: Eigenspectrum Comparison

$$\text{EffRank} = (\|\lambda\|_1 / \|\lambda\|_2)^2$$

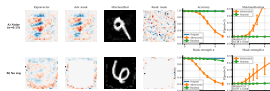


Figure: Adversarial Robustness

Eigenvector-based attacks

Language figures to be filled by Itay

Available figures in Report/figures/language/:

- ▶ `figure_8_negation_circuit.pdf`
- ▶ `figure_9a_correlation_progression.pdf`
- ▶ `figure_9b_correlation_histogram.pdf`

Reproduction Summary

Claim	Description	Status	Notes
<i>Vision (Section 4)</i>			
V1	Low-rank emergence	✓	Eff. rank: 15.5 (WD) vs. 38.5 (none)
V2	Interpretable eigenvectors	✓	Clear digit patterns with regularization
V3	Overfitting without reg.	✓	Diffuse patterns without regularization
V4	Regularization roles	✓	WD reduces rank; noise improves accuracy
V5	Cross-seed consistency	✓	Std < 1% across 5 seeds
V6	Adversarial generation	✓	Eigenvector-based attacks work
<i>Language (Section 5)</i>			
L1	SAE feature discovery	✓	Pretrained SAEs work for analysis
L2	Negation/AND-gate circuits	✓	Opposing eigenvalues confirmed
L3	Low-rank interactions	~	18–32% >0.75 vs. paper's 69%
L4	Sparse interactions	✓	Clear block structure in feature 751
L5	Cross-model generalization	~	Methodology works; magnitudes differ

Extension 1: Background and Purpose

Hypothesis: Eigenvector decomposition generalizes across datasets

Test Cases:

- ▶ **Writer Independence:** MNIST $\leftrightarrow$ EMNIST-Digits
- ▶ **Domain Transfer:** MNIST $\rightarrow$ USPS ($16 \times 16 \rightarrow 28 \times 28$)
- ▶ **Geometric Semantics:** MNIST $\rightarrow$ EMNIST-Letters
 - ▶ $O \rightarrow 0, I \rightarrow 1, Z \rightarrow 2, S \rightarrow 5$

Intuition: Geometric resemblance should lead to similar eigenvectors

All models use Center-of-Mass (CoM) normalization to focus on shape geometry

Extension 1: Results - Generalization Works!

Cross-Dataset Accuracy:

- ▶ MNIST $\rightarrow$ EMNIST-Digits: **97.7%**
- ▶ EMNIST-Digits $\rightarrow$ MNIST: **97.9%**

USPS Transfer:

- ▶ Regularized: **72.3%**
- ▶ Baseline: 53.4%
- ▶ Improvement: **+19 pp**

Letter $\rightarrow$ Digit Mapping:

- ▶ O $\rightarrow$ 0: **99.6%**
- ▶ I $\rightarrow$ 1: **87.1%**
- ▶ Z $\rightarrow$ 2: **88.8%**
- ▶ S $\rightarrow$ 5: **87.4%**

Extension 1: Eigenvector Comparison

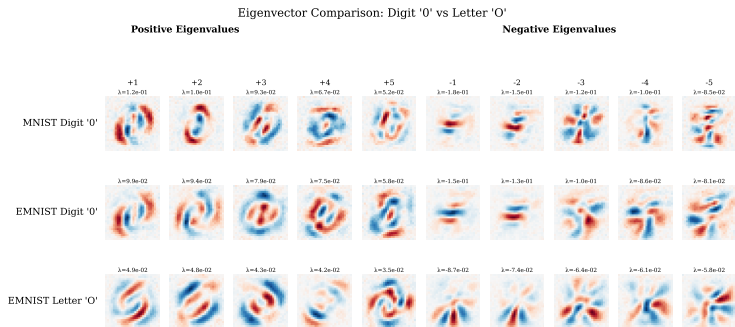


Figure: Eigenvector Comparison: Digit '0' vs Letter 'O'

Quadratic Form Similarity

Problem: Cosine similarity fails

- ▶ Similar pairs (0–0): 0.358
- ▶ Dissimilar pairs (0–X): 0.339
- ▶ Not discriminative!

Solution: Quadratic Form Similarity

$$\text{sim}(A, B) = \frac{\text{tr}(A \cdot B)}{\|A\|_F \|B\|_F} \in [-1, 1]$$

Results:

- ▶ Similar pairs: **0.400**
- ▶ Dissimilar pairs: **-0.060**
- ▶ Large gap with $p < 10^{-4}$

Incorporates eigenvalue signs and eigenvector alignment

Extension 1: Quadratic Form Similarity Heatmap

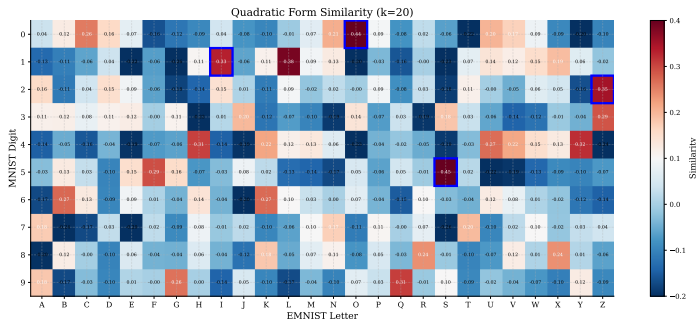


Figure: Quadratic Form Similarity Heatmap

Clear discrimination: strong diagonal for MNIST vs. EMNIST-Digits; bright entries at shape-matched pairs (0-O, 1-I, 2-Z, 5-S)

Extension 1: Key Figures

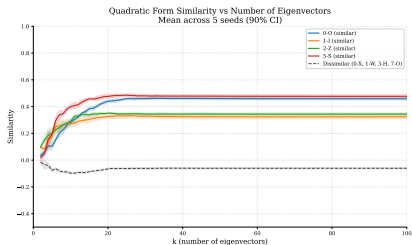


Figure: Similarity vs Top-k Eigenvectors

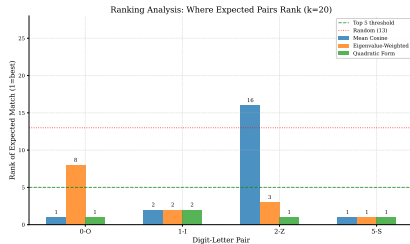


Figure: Ranking Analysis: 3/4 similar pairs rank #1

Extension 2: Background and Purpose

CP-Decomposition: Architectural constraint shifting from **post-hoc** to **intrinsic** interpretability

What it replaces:

- ▶ Original: Bilinear layer computes interaction tensor from W_l and W_r matrices
- ▶ CP: Replaces the bilinear layer structure, directly parameterizing the interaction tensor with rank- R constraint: $W_{cij} = \sum_{r=1}^R \lambda_r A_{cr} B_{ir} C_{jr}$

Maintains interpretability:

- ▶ CP factors can still be eigendecomposed
- ▶ Factors are **discrete parts** and **localized** (e.g., top loops, vertical strokes)
- ▶ More disentangled than dense eigenvectors

Addresses orthogonality limitations:

- ▶ Standard eigendecomposition enforces orthogonality $\rightarrow$ entangled features
- ▶ CP does **not** enforce orthogonality $\rightarrow$ factors naturally overlap
- ▶ It's efficient: reduces parameters from $\mathcal{O}(d_{\text{hidden}} \times d_{\text{in}}^2)$ to $\mathcal{O}(R \times d_{\text{in}})$

Three variants: Fixed CP, Lambda CP (L1), Gated CP (L0)

Extension 2: Results - It Works!

Accuracy vs Rank Tradeoff:

- ▶ Accuracy increases with rank
- ▶ At $R = 256$: **93.8%** accuracy
- ▶ Effective rank: **17.5**

Comparison:

- ▶ Original (dense): 97.5%
- ▶ CP ($R = 256$): 93.8%
- ▶ Drop: 3.7 pp

But: Drastic runtime improvement!

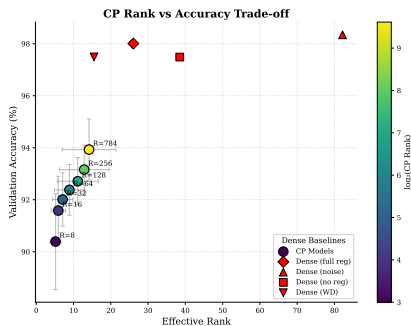


Figure: Accuracy vs Rank Tradeoff

Extension 2: Key Figures

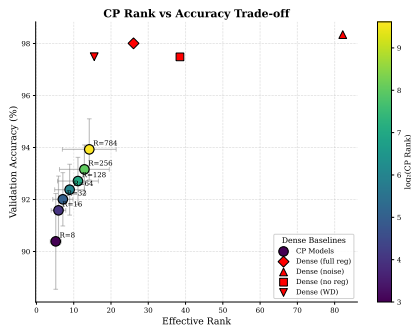


Figure: Accuracy vs Rank Tradeoff

Experiment	Runs	Wall (h)
Vision (Section 4) [†]	81	1.22
Language (Section 5) [‡]	2	0.19
Extension 1	15	0.58
Extension 2 (CP)	105	0.05
Total	203	2.04

Table: Environmental impact (CO₂ via codecarbon)

[†] Some experiments on local M4

[‡] Language experiments on local M4 (MPS)

Extension 2: Disentangled Factors



Conclusion

Main Contributions:

- ▶ **Vision:** Fully reproduced
 - ▶ Effective rank ratio $0.40 < 0.5$ ✓
 - ▶ Interpretable eigenvectors confirmed
- ▶ **Language:** Placeholder (to be filled)
- ▶ **Extension 1:** Cross-dataset robustness
 - ▶ Generalization works! (97.7% cross-dataset accuracy)
 - ▶ Quadratic Form Similarity successfully discriminates
- ▶ **Extension 2:** CP-decomposition
 - ▶ 93.8% accuracy at $R = 256$ with drastic runtime improvement
 - ▶ Disentangled, atomic feature representations

CP-Decomposition Variants

Three variants with different regularization approaches

Fixed CP

Fixed CP: Hard rank constraint with no additional regularization

Characteristics:

- ▶ Strict complexity bound: rank R is fixed during training
- ▶ No regularization on the factors or scaling parameters
- ▶ Operates as a strict bottleneck

Results:

- ▶ Effective rank ~ 3.5 regardless of allocated capacity ($R \in \{32, 64, 128, 256\}$)
- ▶ Accuracy: $\sim 91.9\%$ (vs. 97.5% dense)
- ▶ $\sim 6\%$ accuracy cost for extreme sparsity

Forces extreme sparsity at the cost of accuracy

Lambda CP (L1 Regularization)

Lambda CP: L_1 regularization on learnable scales λ_r

Characteristics:

- ▶ Soft sparsity via L_1 regularization on λ_r parameters
- ▶ Allows factors to be pruned naturally during training
- ▶ More flexible than Fixed CP

Results:

- ▶ At $R = 256$: **93.8%** accuracy
- ▶ Effective rank: **17.5**
- ▶ Comparable to dense models with weight decay (eff. rank 15.5)

Provides robust middle ground: competitive accuracy with structural disentanglement

Gated CP (L0 Proxy)

Gated CP: Hard sigmoid gates with L_0 proxy for explicit rank pruning

Characteristics:

- ▶ Uses L_0 regularization proxy: $\sum_r \sigma(\phi_r)$ approximates count of active ranks
- ▶ Learnable gate logits ϕ_r control continuous gate values (0–1)
- ▶ Hard sigmoid gates: $\text{clamp}(\sigma(\phi_r) \cdot 1.1 - 0.05, 0, 1)$
- ▶ Most flexible regularization approach

Results:

- ▶ At $R = 256$: **93.79%** accuracy
- ▶ Effective rank: **17.35**
- ▶ Similar performance to Lambda CP

Alternative approach to Lambda CP with similar performance; both provide efficient, interpretable modeling